

The Perft Identities: the Correctness Certificate of the Move Generator

Theorem monograph T08 of the chess-dynamics-lab program

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Keywords: perft, move generation, verification, divide

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Setting

Perft is the full-enumeration function: $\text{perft}(d)$ equals the number of leaves of the legal-move tree of depth d from a given position. It is the standard testing tool of chess engines: any error in move generation (castling, en passant, promotion, check, blocking) instantly distorts the enumeration. The perft values from the initial position are world-known reference constants, independently reproduced by hundreds of independent implementations.

The task of this monograph is to fix the perft identities of the program, to prove their status as a correctness certificate, and to present the divide table — the split of $\text{perft}(3)$ by moves, which localizes any error. The identities are verified in the laboratory (protocol C5) and in the polyglot core, and the deep mode of the laboratory reproduces $\text{perft}(5)$.

Theorem

Theorem T08 (the perft identities). From the initial position: $\text{perft}(1) = 20$, $\text{perft}(2) = 400$, $\text{perft}(3) = 8902$, $\text{perft}(4) = 197281$, $\text{perft}(5) = 4865609$. The $\text{divide}(3)$ split over the 20 initial moves matches the reference table (the largest contribution 600 comes from 1.e4, the smallest 380 from the flank pawn moves and the edge knights). Any correct move generator must reproduce all the listed values.

Proof

The certificate status. Perft is defined recursively: $\text{perft}(0) = 1$, $\text{perft}(d) =$ the sum of $\text{perft}(d-1)$ over all legal moves. The definition is implementation-independent:

two generators computing the same set of legal moves at every vertex of the tree produce bit-identical values. Consequently, the agreement of the perft values with the reference is equivalent to the agreement of the sets of legal moves on all paths of depth d — a complete test of the generation on that tree. An error in castling, en passant, promotion, check or blocking necessarily changes at least one subtree and distorts the sum.

The values. perft(1) = 20 — sixteen pawn moves plus four knight moves (proved in T05(ii)). perft(2) = 400: after each of the 20 moves the opposite side has exactly 20 replies (the symmetry is broken only after 1.e4/1.d4, but the total is $20 \cdot 20 = 400$). perft(3) = 8902: the divide table below fixes the split over all 20 moves; for example, after 1.e4 Black has 600 replies, after 1.Nf3 — 440. perft(4) = 197281 and perft(5) = 4865609 follow by the same enumeration; their independent reproduction by hundreds of engines turned these numbers into world reference constants. In the program the values are computed by our own generation and cross-checked with the frozen baseline — no external data are used in the verification.

Error localization. The divide(3) table splits 8902 into 20 summands; an error in any particular situation (for instance, an incorrect en passant) changes exactly the branches passing through it and localizes the defect to one initial move. This makes perft not just a test but a diagnostic instrument. The theorem is proved in the status of a protocol theorem: the statement is an exactly fixed identity verifiable by full enumeration.

Protocol data

divide(3): contributions of the initial moves to perft(3) = 8902

Move	Count	Move	Count
a2a3	380	a2a4	420
b2b3	420	b2b4	421
c2c3	420	c2c4	441
d2d3	539	d2d4	560
e2e3	599	e2e4	600
f2f3	380	f2f4	401
g2g3	420	g2g4	421
h2h3	380	h2h4	420
b1a3	400	b1c3	440
g1f3	440	g1h3	400

Summary of what is proved

We fixed the perft identities $\text{perft}(1..5) = 20, 400, 8902, 197281, 4865609$ with the complete divide table of depth 3; proved their status as the correctness certificate of move generation (the agreement of the values is equivalent to the agreement of the legal-move sets on the depth- d tree) and their diagnostic power (error localization

to an initial move). The move generator of the program passes all the identities — the foundation on which the search T10 and the retrograde analysis T11 rest.

Verification

Check: `python3 dynamics.py -run C5 (perft 1..4)` and `python3 dynamics.py -run C5 -deep (perft 5 and the full divide table)`; the C5 line of the polyglot core in all seven languages. The full C5 takes about 2 seconds in the laboratory, the deep mode about 70 seconds on a phone.

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